

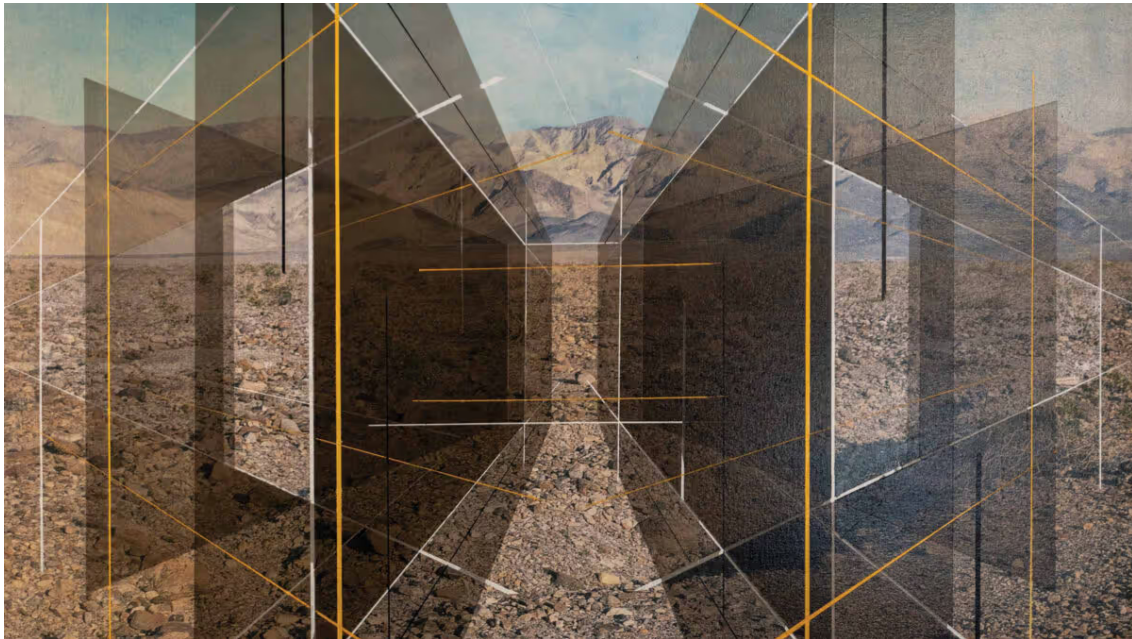


Organizational Decision Making

How AI Agents Orchestrate Work Across Silos

by Kris Johnson Ferreira and Jordan Tong

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Rodrigo Valenzuela

Summary. While many companies now use AI to improve individual tasks, most still struggle to apply it to the cross-functional work that makes or breaks enterprise performance. Drawing on field research at companies including Walmart, Amazon, Ericsson, Ramp, and... [more](#)



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Imagine a new tariff is announced on imported goods that are crucial to a company's business and will take effect in 60 days. The CEO of the company recommends "pulling forward" inventory—buying it sooner than originally intended and importing it before the tariff hits—to protect margins and market share. The logic is clear. Yet weeks later the move hasn't been carried out as the CEO wanted. After rounds of revisions and compromises among different functions—including procurement, logistics, finance, legal, marketing, sales, and merchandising—the company implements a patchwork of suboptimal actions that fail to secure a large portion of the crucial goods before the tariff goes into effect.

Over the past decade, companies have invested heavily in smarter tools, including AI assistants that draft documents and summarize reports and machine-learning systems that forecast demand. When employed well, they improve the performance of clearly defined, narrowly scoped tasks within functions. But most companies have yet to address a deeper structural challenge: how to employ AI to make and implement enterprisewide decisions. This is the next frontier: AI-human systems that coordinate decisions and knowledge across functions.

These are what we call *agentic AI orchestration systems*, and in our field research on human-AI collaboration, we've interviewed executives at Walmart, Amazon, Ericsson, Ramp, Medtronic, and other companies that are building them. They're making real progress, but the initiatives are still in the early stages: Even the most advanced organizations have far to go to achieve full enterprisewide orchestration. Still, it's possible to see how this technology will evolve, and our additional ongoing research on the most effective ways for AI and humans to work together reveals how these systems should be set up. We consistently find that people create the most value not by competing with AI but by

contributing information it does not have—such as tacit knowledge and updates about changing constraints and contextual shifts. The challenge, then, is not deciding whether humans or AI should make decisions. It is designing systems that combine the strengths of both.

Realizing orchestration systems' potential will require more than installing new technology. Companies will have to redesign how higher-level decisions flow from lower-level ones, how human-AI interactions are structured, and how workflows are integrated. The payoff will be substantial: higher-quality cross-functional decisions that are executed faster without surrendering human judgment or control. In fact, these systems will preserve—and in some cases strengthen—managerial authority by embedding it in the ways decisions are structured, informed, and governed. Organizations that build AI orchestration capability into their infrastructure—rather than relying on managers to stitch things together—will gain an advantage over their rivals.

The Hidden Bottleneck

The decisions that matter most in modern organizations typically span multiple functions. They involve interacting constraints, data from multiple sources, and human knowledge and judgment shaped by experience rather than spreadsheets—which often must be applied in dynamic situations where outcomes and downstream effects are inherently uncertain.

Tariff responses are just one example. The same challenges often emerge when companies experience a sudden, unexpected event that significantly increases or decreases demand for their offerings, suffer supply chain disruptions, or redesign their product portfolios. These kinds of situations require enterprisewide solutions, whose quality depends on how well the organization integrates partial, distributed knowledge into a coherent course of action.



Acting on the decision to pull forward inventory before a tariff takes effect, for example, requires addressing a web of interdependent questions: How will customers respond to price changes? To what extent are suppliers willing to accelerate production? Do our warehouses have the capacity to hold additional inventory? Can our transportation networks accommodate a surge of deliveries? Are there product-life-cycle or obsolescence risks that make front-loading inventory unattractive for some SKUs? How will working capital, risk exposure, and earnings volatility change under different scenarios?

Most organizations have the teams, tools, data, and expertise needed to address such questions. But many aren't skilled at gathering information from disparate parts of the organization and synthesizing it quickly. That task typically falls to a senior leader, who must decide which questions to ask, anticipate constraints that haven't yet surfaced, reconcile conflicting objectives, and manage iterations as new information emerges. Much of that information exists only in people's heads. It surfaces informally, unevenly, and often too late. A logistics manager mentions that warehouse capacity and labor are already committed for a seasonal peak. A procurement leader points out that a key supplier is willing to speed up deliveries only if volumes are guaranteed across multiple product lines. Each late discovery forces the process to reset. Scenarios are reworked. Assumptions are revised. Meetings multiply.

It's tempting to blame such breakdowns on culture—on bureaucracy, risk aversion, or misaligned incentives. But the fundamental causes are these: No single individual's brain can hold all the relevant information, constraints, and scenarios. Information is asymmetric; each function knows things that others do not, and often people cannot articulate what will matter until it matters. Issues and risks don't show up all at once; they emerge one by one. So delays in carrying out strategies that entail a host of cross-functional decisions aren't signs of managerial

a host of cross-functional decisions aren't signs of managerial failure. They're signals that the organization has exceeded its capacity to orchestrate decisions.

Why Better AI Tools Haven't Solved the Problem

It's natural to assume that artificial intelligence should provide relief. After all, many of the tasks that slow cross-functional decisions—forecasting demand, estimating risk, generating documents, extracting information from contracts—are precisely the domains where AI has made impressive strides.

Agentic AI can outperform humans when tasks are narrow and well-defined, inputs are complete, and historical data is representative of the current situation. But orchestrating cross-functional decisions means coping with situations in which those conditions often aren't present.

The solution is to combine human judgment and machine intelligence—both for individual tasks and across the broader decision process. At the task level, this means structuring human-AI collaboration so that people contribute information the system lacks and use it to shape outputs. At the orchestration level, it means connecting those outputs—along with the human knowledge embedded within them—so that they're validated, routed, and integrated across tasks and functions.

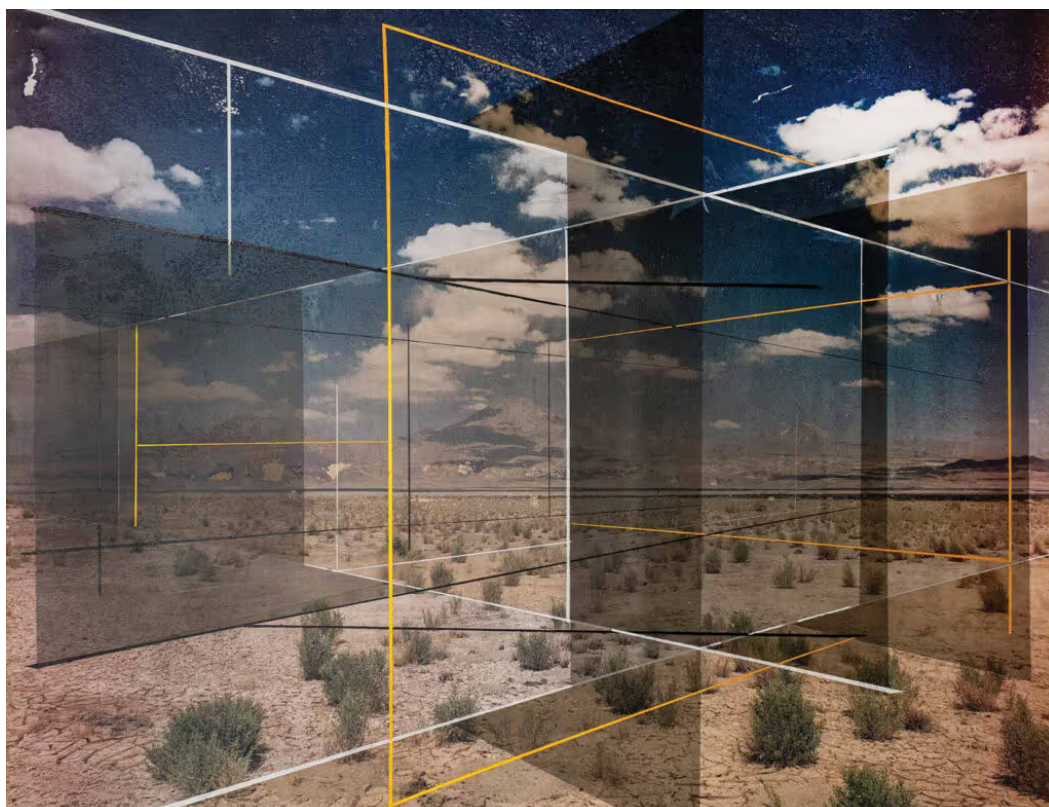
The Task Level

In most organizations today, employees using task-based agentic AI are expected to improve its performance by monitoring and overriding it when they think they can do better. Yet in practice they often struggle to determine when and how to intervene. In our experimental research, we found that humans tend to develop a general level of trust in AI that they then apply to all situations, rather than adjusting their confidence in AI when its performance is likely to be strong or weak. That sometimes causes people to override AI recommendations when they shouldn't or miss



instances when they should, leading to significantly worse outcomes in both cases.

Underlying this pattern is a flawed “human versus AI” mental model. The correct model is “human + AI.” The right approach is to ask, “Do I know something important—context, constraints, or tacit knowledge—that the AI can’t access or infer on its own?” Only when the answer is yes is human intervention valuable.



Rodrigo Valenzuela's *New Land* paintings explore the history of westward expansion, using a labor-intensive printing process to represent the bureaucratic burdens endured by immigrants.

In the tariff example, this kind of human information includes tacit knowledge of supplier concessions or constraints and insights into how demand might change if store shelves are flooded with extra inventory. Such information is often sparse, subjective, and unevenly distributed across the organization. Most task-based agentic AI tools aren't designed to surface or integrate it systematically.

Our research points to a more effective approach. In a series of controlled experiments, we found that performance improves when humans are trained to intervene only when they hold

important information the AI lacks—either supplying it as an input before the system makes its recommendation or using it to make an adjustment afterward—rather than asked to form independent judgments that compete with the AI's.

Practically, the goal is to have people address questions like these: Does the AI have all the critical input data? Can I still trust the historical relationships between the inputs and the outcome? Is this situation represented in the data the model learned from? If the answer to all those questions is yes, people need not intervene. If the answer to any of them is no, people should adjust the AI tool's inputs or output to incorporate the human-only information. This can be accomplished by creating a simple checklist that employees can review when using task-based AI agents or by building an AI agent that prompts employees to ask these questions and suggests how to integrate the answers into the tool.

In the hypothetical tariff example, here's how human-AI collaboration could work:

- A procurement large language model (LLM) scans supplier contracts and identifies excess capacity that could be tapped to speed up the delivery of orders before the tariff goes into effect. But a purchasing manager has had a recent conversation with a key supplier that has revealed critical information the system lacks: Several of the supplier's other customers are pursuing the same strategy, and the supplier will dedicate additional capacity to the company only if it expands its purchases to other items it doesn't buy from the supplier and sell in its stores. In response, the procurement manager revises the LLM's inputs or output to reflect that what appears feasible in the contracts is actually contingent on awarding the supplier additional purchase orders for items the company doesn't currently sell.

- An AI tool has used data on early-season sales of the current product assortment to predict demand for it later in the season. A merchandiser, however, recognizes that early-season sales of the subsequent assortment would be adversely affected by overstocking current products to mitigate the impact of a higher tariff and therefore adjusts the AI's demand forecast to take that into account.
- When reviewing AI forecasts for warehouse capacity utilization, a logistics manager recognizes that the relationship between inputs and outputs has shifted since the model was trained: The head of the seasonal-products division has decided to double the amount of in-store inventory for the upcoming Christmas season to support a new product launch. In response, the manager revises the AI's baseline forecast upward to reflect the expected increase and sees that there wouldn't be enough capacity to house extra inventory pulled forward in anticipation of the tariff.

This is a fundamentally different workflow. The responsibility of humans is not to form beliefs about what the best decision should be. Their job is to identify whether they possess information the task-based AI agent doesn't and, if they do, to revise the tool's inputs or outputs to incorporate that knowledge.

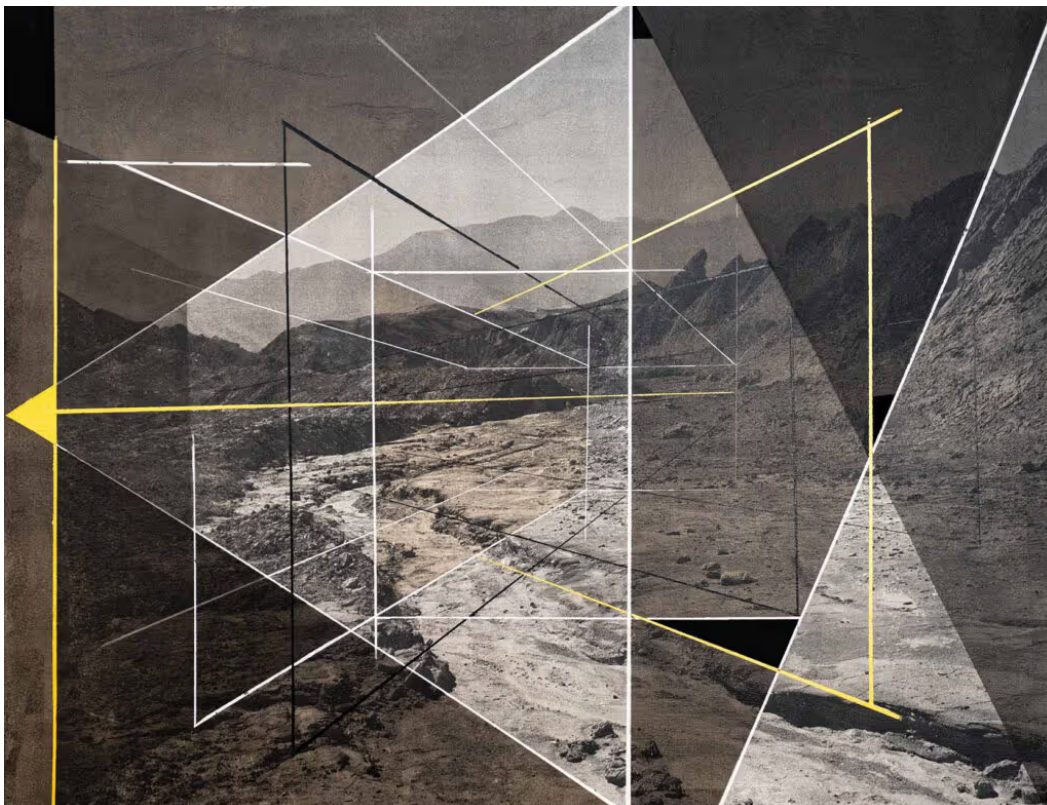
The Orchestration Level

The most consequential decisions aren't made within a single task; they emerge from linking together many narrowly scoped tasks. That explains why, despite significant investment in AI and even with better human-AI collaboration, organizations still struggle to move faster on their most important decisions. What's been missing is a system that ensures that human-only information and the outputs of individual tasks are validated and shared with other parts of the organization.

At the foundation of this system is a rule-based layer of agents we call *connectors*. They take outputs from task-based AI agents, verify that those outputs stay within management-specified guardrails, trigger the next step in the workflow, and pass relevant information on to other task-based agents as inputs or to a higher-level orchestrator. The role of connectors isn't to make decisions; it's to ensure that multitask or cross-functional workflows are executed effectively.

The connectors' guardrails let managers set business rules and decide how much freedom the system has when making decisions. These can be simple—like checking that something is feasible or making sure results fall within an expected range. They can also be more advanced, using other AI tools to run additional checks. For example, a connector might double-check that a forecast isn't unusually high or low according to limits set by management. If something doesn't meet the rules, the connector can stop the process, redirect it, or ask a conversational AI agent to get more input from an employee. In sum, connectors help information move through the system in a structured, consistent, and reliable way.

Overseeing this layer is the *master AI orchestrator*: an AI agent that employees engage with to share information only they possess, describe external issues that may affect the business (like a new tariff), surface potential opportunities (like pulling forward inventory), and explore what-if scenarios. The master orchestrator doesn't make high-level enterprise decisions; it structures and coordinates the analyses and information needed for humans to make them. In conversational exchanges, it gets employees to share human-only knowledge and articulate objectives, constraints, and decision boundaries. It helps identify and coordinate which tasks, workflows, and employees need to be involved in a decision, routes questions to the appropriate tools and people, and manages the overall decision process.



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The master AI orchestrator creates a plan, and the connectors execute it by triggering and advancing workflows, validating outputs, and routing information across tasks in a disciplined and repeatable way. Together, the orchestrator and the connectors create a continuous flow of information in which task outputs and critical human-only knowledge become structured inputs for other tasks, vastly improving coordination and drastically shrinking the time needed to make and implement decisions.

Let's return to the tariff example. In a traditional organization, the decision to pull forward inventory would trigger a chain reaction of emails, meetings, requests for analyses, and discovery of constraints. Even with task-based AI agents, the burden of stitching together information would still fall on a human orchestrator, and critical information would often arrive late—or never travel beyond the silo where it resides.

With agentic AI orchestration, the process unfolds differently. The CEO identifies the tariff issue and the potential opportunity to pull forward inventory and enters this information into the

master AI orchestrator, which asks the CEO to clarify objectives and constraints. It then engages the relevant workflows and functional managers—in demand planning, procurement, logistics, merchandising, and finance—and initiates parallel analyses using task-based AI agents. As these analyses run, connectors link them together and prompt the master AI orchestrator to elicit and share relevant human-only knowledge.

Let's look at what happens with the procurement workflow. The master AI orchestrator alerts the procurement manager of the CEO's desire to explore pulling forward inventory and triggers the procurement contract-scanning LLM to identify suppliers with possible excess capacity. This task-based tool generates a list of relevant suppliers and contract terms, and a connector provides that output to the master AI orchestrator, which prompts the procurement manager for critical human-only information. After holding discussions with the potential suppliers, the procurement manager might respond that one supplier's excess capacity is available only if the company commits to buying additional items from that supplier. The moment this information is shared with the master AI orchestrator, it's immediately disseminated to all relevant task-based AI agents and employees. Merchandising is alerted to the implications for the breadth of the company's product assortment and shelf space, and connectors trigger its task-based forecasting agents to take the new information into account and update demand scenarios. Finance is also alerted about the conditional supplier capacity, and connectors automatically trigger task-based finance agents to recalculate the potential impact on working capital and earnings volatility. The final decision on whether to pursue the pull-forward strategy—and on what scale—is made by the senior leaders, such as the heads of supply chain, merchandising, and finance, who now have a fully integrated view of the trade-offs.

The effect is twofold. Speed increases because workflows proceed in parallel and iteration is no longer driven by late surprises.

Quality improves because each task incorporates validated inputs from elsewhere in the system.

Notably, the orchestration system prevents task-based agents from working at cross-purposes. Last June data science leaders at Walmart, which is in the process of building such an orchestration system, told us that specialized agents can generate high-quality recommendations and human-in-the-loop actions tailored to their respective domains. However, when those agents operate independently, their outputs may conflict. The orchestration layer helps reconcile those tensions, ensuring that agents' outputs remain aligned with enterprise priorities and contribute to coherent business outcomes.

Agentic AI orchestration thus functions as a shared organizational infrastructure—one that's activated by questions and opportunities raised by leadership. Authority and accountability remain firmly with humans. Enterprise decisions aren't automated; what's automated is the orchestration of intelligence, including the routing of questions, the integration of insights, and the continuous alignment of local analyses with enterprise objectives.

Building Toward Agentic Orchestration

Organizations cannot use AI to break coordination bottlenecks if their underlying decisions are opaque, entangled, and idiosyncratic. So the first thing leaders must do is make sure decision processes are defined and structured well. That effort should proceed from the bottom up, and it won't succeed unless the entire organization commits to it. Only the top leadership can make that happen.

Step 1: Break decisions down into modules. Two companies we studied—a global e-commerce giant and Xsight, an online platform that helps manufacturers sell into overseas markets—found that orchestration improved significantly when decisions



were divided into precise, well-bounded tasks. This entails specifying each task's inputs, outputs, constraints, and objectives. The scope of each task should be narrow enough that it can be easily understood, evaluated, and linked to other tasks. Information and assumptions should be standardized; for example, procurement, logistics, and finance should all work from the same definition of supplier capacity, whether it reflects theoretical production capacity, contractually committed capacity, or realistically achievable capacity.

Step 2: Strengthen task-based agentic AI. Begin applying agentic AI to the tasks that are largely driven by data that AI can access and where the expected returns are the greatest. In every case the collaboration between people and the AI should be structured. The AI should be designed to make its inputs, assumptions, and reasoning visible to the people involved in the task and to actively elicit the relevant knowledge that only they hold. For example, the system might ask the procurement manager whether he had recent supplier conversations that might affect what the model treats as available capacity. On the human side, people must understand what the AI can and cannot see so that they can recognize when circumstances have changed, when assumptions no longer hold, and when they possess knowledge the system has no way of inferring—and provide that information to the AI or, when needed, override the AI. To this end, people must be trained to ask, “Is the AI system missing something I know?” rather than “Do I agree with the AI system?”

Making this work requires a cultural shift. Everyone needs to understand that AI acts as a structured partner in decision-making. Leaders—from the top to the functional-team level—have to reinforce that mindset.

Step 3: Add the master orchestration. Once task-based AI agents and their human collaborators achieve a satisfactory level of performance, your company can take the next step to orchestrate their collective work. Start by coordinating closely related tasks



within individual small groups. Build simple connectors linking one task to the next, and develop an initial master AI orchestrator to interface with the employees involved in those tasks. This will support an agile approach to developing the orchestration software—one that emphasizes iterative progress, rapid feedback, collaboration, and adaptability. When the orchestrator matures (becoming more developed, capable, stable, and reliable) and works effectively for several groups of closely related tasks, it should be extended across more modules.

In multiple industries companies are beginning to build early versions of such infrastructure. At the fintech platform Ramp and at the healthcare technology company Medtronic, core decision processes have been organized into distinct tasks, each governed by a separate orchestrator. Ramp is now developing a single enterprise master AI orchestrator that will route employee queries across the modular systems and engage the appropriate agents as needed.

Step 4: Manage the system's evolution. Agentic AI orchestration is not a system you build and leave alone. Leaders must continually ask, “Are modules defined narrowly enough for decisions to be orchestrated effectively? Is human-only information being captured in a way that makes it accessible across the system? Are incentives aligned to encourage early disclosure of constraints rather than late-stage escalation? Do employees trust and understand AI well enough to collaborate with it rather than compete against it?” The answers will change as the technology and the business’s capabilities, constraints, and objectives change—and leaders must update roles, governance, and training accordingly. They should see themselves as the architects of a living decision infrastructure—one that improves as the organization and its AI capabilities mature together.

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Agentic AI orchestration should not be viewed as a means to



totally automate decision-making. The goal should be to automate the orchestration of intelligence—the ability to route questions, integrate insights, and align local analyses with enterprise objectives in real time. Organizations that build this capability will compound the value of every piece of human knowledge across the enterprise while their rivals are still discovering critical constraints in the third round of alignment meetings.

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